Distributed Streaming Anomaly Detection on Blockchain Transaction Graphs

Review 1:

Faculty Name: Dr Banu Priya Prathaban

Team Members:

RAYAAN ARORA - RA2411031010073

FAHEEM MUSHTAQ - RA2411031010079

LAKSHITA KUMARKAR - RA2411031010091

AARUSHI SARKAR - RA2411031010099

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Introduction

- Blockchain networks generate a continuously growing stream of transactions that can be represented as **dynamic transaction graphs**.
- Each transaction can be represented as a **node**, while the movement of funds between transactions can be represented as **directed edges**.
- Although blockchain transactions are publicly recorded, illicit activities such as fraud, ransomware, scams and money laundering can be difficult to identify because suspicious behaviour evolves over time.
- Existing blockchain anomaly-detection approaches largely analyse **static or pre-constructed temporal datasets**, rather than continuously updating the analysis as new blockchain blocks arrive.
- This project proposes a **distributed streaming graph analytics framework** for continuously processing blockchain transactions and identifying potentially anomalous transaction patterns.

Motivation

- **Why this problem?**
- Blockchain transaction graphs continuously grow as new blocks are generated.
- A static analysis may become outdated as new transactions change the structure and behaviour of the network.
- Illicit transactions represent a relatively small fraction of overall activity, making detection a highly imbalanced problem.
- Processing large and continuously evolving transaction graphs requires **distributed Big Data processing**.
- Near-real-time anomaly detection can potentially support faster financial-crime monitoring and investigation.
- **Sustainable Development Goals**
- **Primary: SDG 16 – Peace, Justice and Strong Institutions**
- Supports transparency, accountability and safer financial systems.
- Helps identify potentially illicit financial activity.
- **Secondary: SDG 9 – Industry, Innovation and Infrastructure**
- Applies scalable distributed computing to large-scale digital infrastructure.

Literature Review

Author / Publication	Title	Contribution	Inference
Weber et al., KDD Workshop (2019)	Anti-Money Laundering in Bitcoin: Experimenting with Graph Convolutional Networks for Financial Forensics	Introduced the Elliptic Bitcoin transaction dataset and evaluated LR, RF, MLP and GCN for illicit-transaction detection.	Established graph-based learning as a useful approach for Bitcoin AML; dataset provides a strong benchmark for our work
Pareja et al., AAAI (2020)	EvolveGCN: Evolving Graph Convolutional Networks for Dynamic Graphs	Proposed a GCN architecture that evolves its parameters with changing graph structure and time.	Demonstrates the importance of modelling evolving graphs; motivates our streaming graph approach..
Li et al., IET Blockchain (2023)	BlockDetective: A GCN-Based Student–Teacher Framework for Blockchain Anomaly Detection	Used a GCN-based student–teacher framework to detect fraudulent cryptocurrency transactions.	Improves anomaly detection and reduces model complexity, but further work is needed for real-time processing.
Wan & Li, Symmetry (2024)	A Novel Money Laundering Prediction Model Based on a Dynamic Graph Convolutional Neural Network and Long Short-Term Memory	Combined dynamic GCN and LSTM to model spatial and temporal transaction relationships.	Shows the value of temporal graph modelling, but uses dynamic snapshots rather than continuous streaming.

Literature Review

Author / Publication	Title	Contribution	Inference
Chen et al., Advanced Intelligent Systems (2025)	Multi-Distance Spatial-Temporal Graph Neural Network for Anomaly Detection in Blockchain Transactions	Proposed MDST-GNN combining multi-distance graph convolution, temporal modelling and self-supervised learning for blockchain anomaly detection.	Shows that combining spatial and temporal graph information improves detection, but evaluation remains dataset-based rather than continuous streaming
Chang et al., Sensors (2025)	Anomalous Node Detection in Blockchain Networks Based on Graph Neural Networks	Proposed SGAT-BC using Graph Attention Networks with subtree attention and ensemble learning to handle highly imbalanced blockchain anomaly data.	Demonstrates effective graph-based anomaly detection, but focuses on offline datasets rather than continuously evolving transaction graphs.
Asiri & Somasundaram, Scientific Reports (2025)	Graph Convolution Network for Fraud Detection in Bitcoin Transactions	Applied Graph Convolutional Networks to identify fraudulent Bitcoin transactions from transaction-graph information.	Confirms the usefulness of GCN-based graph representations for Bitcoin fraud detection, while leaving scope for scalable streaming analysis.
Lin et al., Scientific Reports (2026)	Detecting Illicit Transactions in Bitcoin: A Wavelet-Temporal Graph Transformer Approach for Anti-Money Laundering	Proposed ChronoWave-GNN combining wavelet-based frequency decomposition with temporal graph learning to capture multi-scale transaction patterns.	Highlights the importance of modelling complex temporal behaviour, but remains focused on model-level detection using benchmark data.

Literature Review

S. No.	Author, Title & Year	Objective	Methodology	Significance
1	Yang, Liu & Li, "Anti-Money Laundering Supervision by Intelligent Algorithm," 2023	Detect explicit and implicit money-laundering activity in virtual-currency transactions.	Combined heuristic rules, LSTM + GCN and ensemble anomaly-detection methods.	Demonstrates that combining graph, temporal and anomaly-detection methods can improve AML detection, but does not address continuous distributed graph processing.
2	Han et al., "Illicit Bitcoin Transaction Detection via Feature-Gated Temporal Graph Learning," 2026	Improve illicit-transaction detection under changing temporal distributions.	Feature-gated temporal graph learning using temporal Bitcoin transaction data.	Shows the importance of handling **temporal distribution shifts** in blockchain fraud detection.
3	Lv & Ding, "Phishing Detection on Ethereum via Transaction Subgraphs Embedding," 2023	Detect phishing addresses in Ethereum transaction networks.	Transaction-subgraph embedding incorporating block height, timestamp and transaction amount.	Shows that transaction subgraph structure and temporal features can identify suspicious blockchain addresses.

Gaps Identified

- **Static / Offline Analysis**
- Most existing blockchain anomaly-detection studies operate on **pre-collected datasets or temporal snapshots**, rather than continuously processing newly arriving blockchain blocks.
- **Limited Streaming Graph Processing**
- Existing work focuses heavily on improving the **detection model**, while comparatively less emphasis is placed on designing and evaluating a distributed pipeline that continuously updates transaction graphs as new blocks arrive.
- **Scalability of Evolving Transaction Graphs**
- Blockchain graphs continuously grow in size. There is a need to evaluate how **processing latency, throughput, memory consumption and graph-update cost** change as transaction volume increases.
- **Detection Latency**
- Static analysis may identify suspicious activity only after a dataset is collected and processed. A streaming framework can investigate whether anomaly scores can be updated with **lower detection latency**.
- **Static vs. Streaming Comparison**
- There is a need for an empirical comparison between **conventional batch/static graph analysis and continuously updated streaming graph analysis** under increasing data volumes.

Problem Statement

- Existing blockchain anomaly-detection approaches primarily rely on static or pre-constructed transaction graphs, limiting their ability to continuously respond to the evolving structure of blockchain networks. This project proposes a distributed streaming graph analytics framework using Apache Spark and streaming technologies to incrementally process incoming blockchain transactions, update transaction graphs, and identify potentially anomalous or illicit activity. The study will compare streaming and static approaches in terms of detection performance, latency, throughput, resource consumption and scalability as transaction volume increases.

Dataset

- **Data Sources**
- **1. Bitcoin Blockchain**
- Public, continuously growing blockchain transaction data
- Used for **large-scale and streaming graph analysis**
- Contains historical blocks and transactions
- **2. Elliptic Bitcoin Dataset**
- Labelled Bitcoin transaction-graph benchmark
- **203,769 transactions**
- **234,355 directed edges**
- **49 temporal time steps**
- **166 transaction features**
- Labels: **Licit / Illicit / Unknown**

Key Features

- Transaction ID
- Block Height
- Timestamp
- Transaction Value
- Input & Output Relationships
- Transaction Graph Structure
- Temporal Transaction Patterns
- Licit / Illicit Label

Dataset Role

Bitcoin Blockchain → Streaming + scalability experiments

Elliptic → Model training + validation + ground-truth evaluation

Plan of Action / Member Tasks

Member	Task	Status
Lakshita Kumarkar	Literature review; blockchain AML research; Elliptic dataset analysis	Completed
Aarushi Sarkar	Dataset collection; Bitcoin block/transaction preprocessing	In Progress
Faheem Mushtaq	Spark + Kafka streaming pipeline and distributed graph construction	Planned
Rayaan Arora	Anomaly detection model, evaluation and scalability experiments	Planned

References

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- **[2]** A. Pareja *et al.*, “EvolveGCN: Evolving Graph Convolutional Networks for Dynamic Graphs,” *Proc. AAAI*, 2020.
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- **[5]** “A Novel Money Laundering Prediction Model Based on a Dynamic Graph Convolutional Neural Network and Long Short-Term Memory,” *Symmetry*, 2024.
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Thank you